NOTE I have described filters as being attributes, but that's not always the case. For example, the action filter is called `ResponseCacheFilter`, but this type is `internal` to the ASP.NET Core framework. To apply the filter, you use the public `[ResponseCache]` attribute instead, and the framework automatically configures the `ResponseCacheFilter` as appropriate. This separation between attribute and filter is largely an artifact of the internal design, but it can be useful, as you'll see in section 22.3.

Response caching vs. output caching

Caching is a broad topic that aims to improve the performance of an application over the naive approach. But caching can also make debugging issues difficult and may even be undesirable in some situations. Consequently, I often apply `ResponseCacheFilter` to my action methods to set HTTP caching headers that disable caching! You can read about this and other approaches to caching in Microsoft's "Response caching in ASP.NET Core" documentation at http://mng.bz/2eGd.

Note that the `ResponseCacheFilter` applies cache control headers only to your outgoing responses; it doesn't cache the response on the server. These headers tell the client (such as a browser) whether it can skip sending a request and reuse the response. If you have relatively static endpoints, this can massively reduce the load on your app.

This is different from output caching, introduced in .NET 7. Output caching involves storing a generated response on the server and reusing it for subsequent requests. In the simplest case, the response is stored in memory and reused for appropriate requests, but you can configure ASP.NET Core to store the output elsewhere, such as a database.

Output caching is generally more configurable than response caching, as you can choose exactly what to cache and when to invalidate it, but it is also much more resource-heavy. For details on how to enable output caching for an endpoint, see the documentation at http://mng.bz/Bmlv.

The real power of action filters comes when you build filters tailored to your own apps by extracting common code from your action methods. To